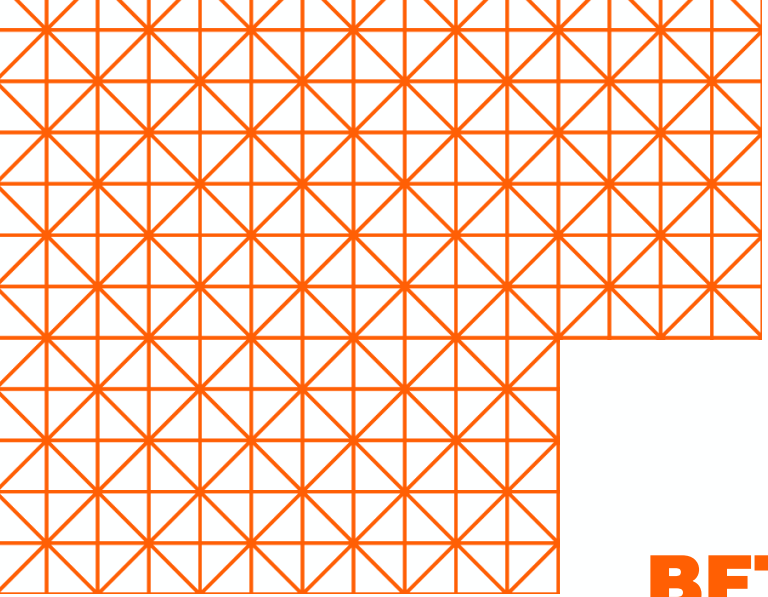




BFT Consensus under Realistic Network Adversaries

Granular partial synchrony with budget link degradation

I ILLINOIS

Vaibhav Mattoo
CS 563

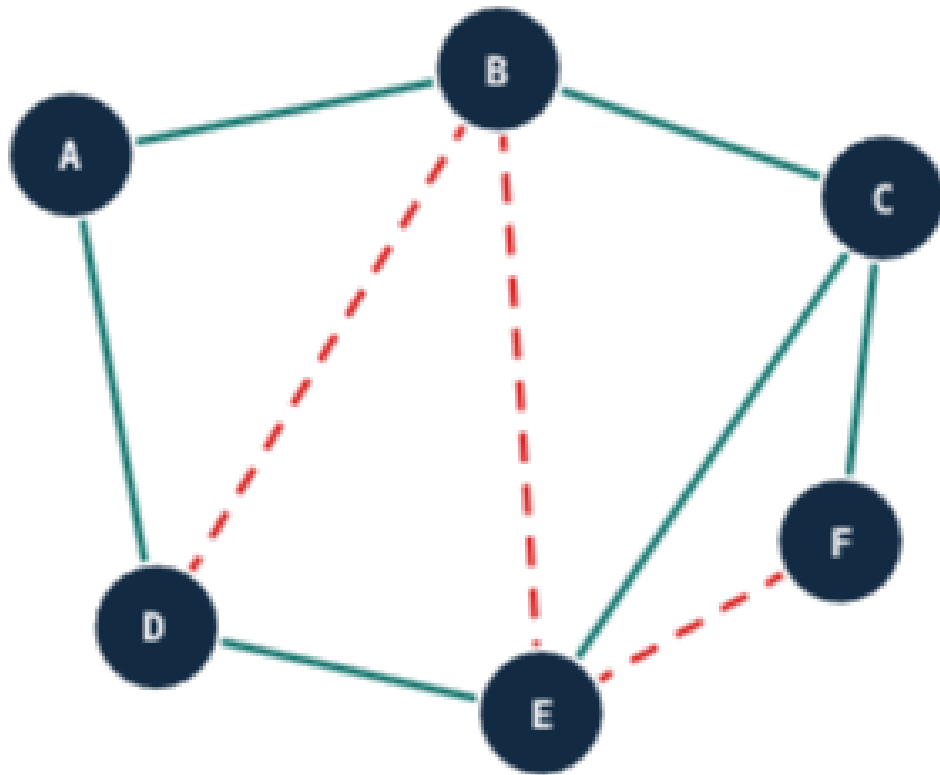
BFT consensus

- **Consensus:** n parties, each party has input v_i , a protocol that solves consensus has 3 properties:
 - Agreement: no two honest parties *decide* on different values.
 - Validity: if all honest parties have the same input value v , then the decision value must be v .
 - Termination: all honest parties must eventually *decide* on some value in the inputs and terminate.
- Consensus is trivial, if all parties are honest and the system is synchronous. To make it non trivial we must fix:
 - Communication model
 - Threshold of the adversary
 - Power of the adversary (also adaptivity, visibility and mobility)
 - Models here are Passive (honest but curious), Crash, Omission and **Byzantine**.
- BFT consensus allows a decentralized network to maintain a single trusted version of the truth, making sure that all honest nodes agree on a valid state even when some nodes act maliciously. Important for banks, air traffic control, blockchains etc.

Communication model

- Asynchrony: adversary can delay messages by any finite amount of time with no specific bound. **FLP impossibility**
- Synchrony: there exists some known finite time bound Δ after which the message must be delivered. **Too unrealistic**
- Partial synchrony: there is a known finite time bound Δ and a special event called GST which the adversary must let happen after some unknown amount of time so that messages sent at time x are delivered by time $\Delta + \max(x, \text{GST})$ **Too coarse**
- **Granular Synchrony [DISC'24]**: network consists of graph consisting of a mixture of synchronous, partially synchronous and asynchronous links.

Granular partial synchrony



Edge types

 **Synchronous:** always $\leq \Delta$

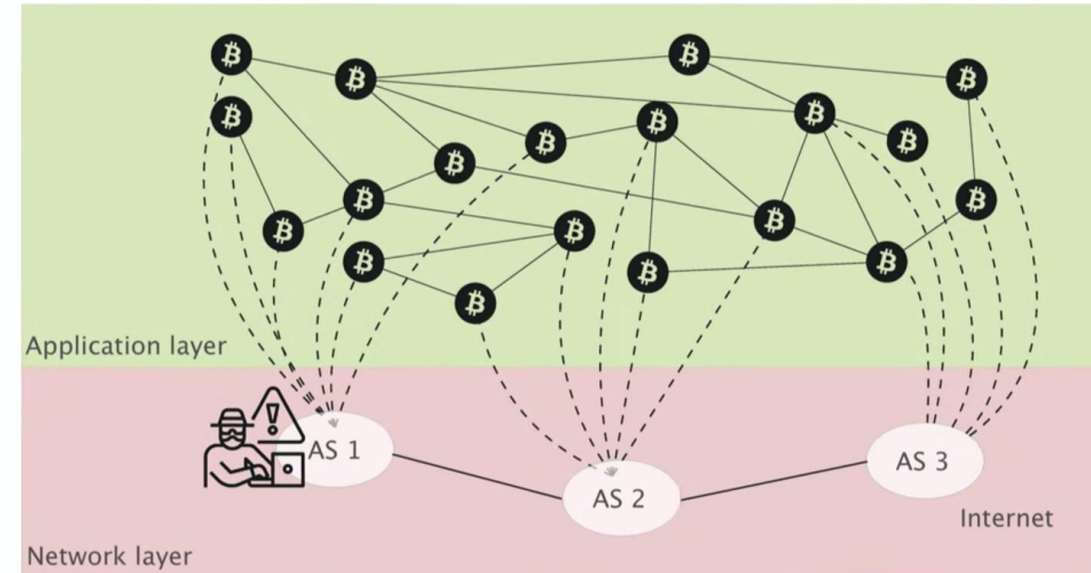
 **Partially sync:** $\leq \Delta$ only after some GST

Prior result:

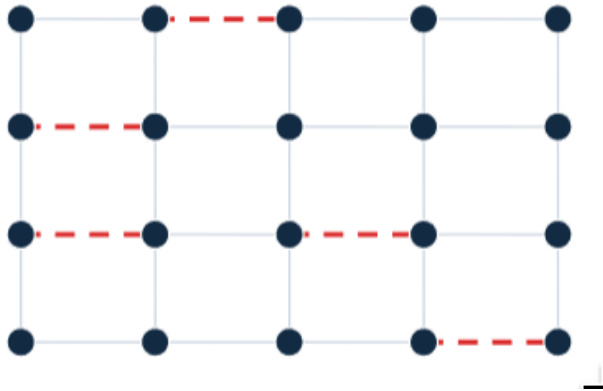
- Under granular partial synchrony, BFT consensus with $n \geq 2f + 1$ on a graph G is solvable if and only if, for any set F of at most f faulty nodes, $\forall A \subseteq V - F$ with $|A| \geq n - 2f$, $\exists B \subseteq V - F$ with $|B| \geq f + 1$ such that $A \rightarrow B$.

The budgeted adversary

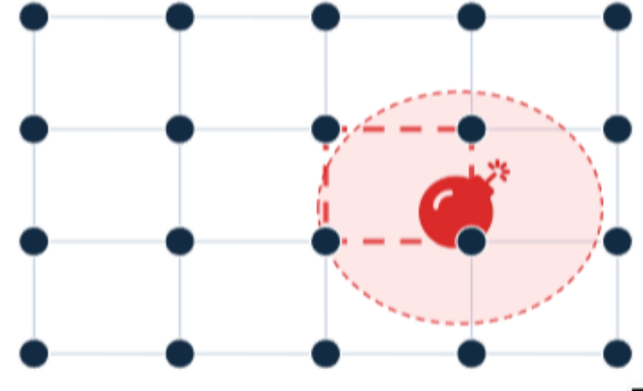
- Degrading a link in the real world takes real resources.
 - Volumetric flooding: Sustain attack traffic on each link you want to flood
 - BGP hijack: control traffic in an AS
- Treat the adversary's degradation power as a budget β . Adversary picks at most β links to degrade and at most f byzantine nodes.
- Questions:
 - 1) Which networks survive every budget β attack?
 - 2) How much slower does the protocol get as β grows?



Models and results

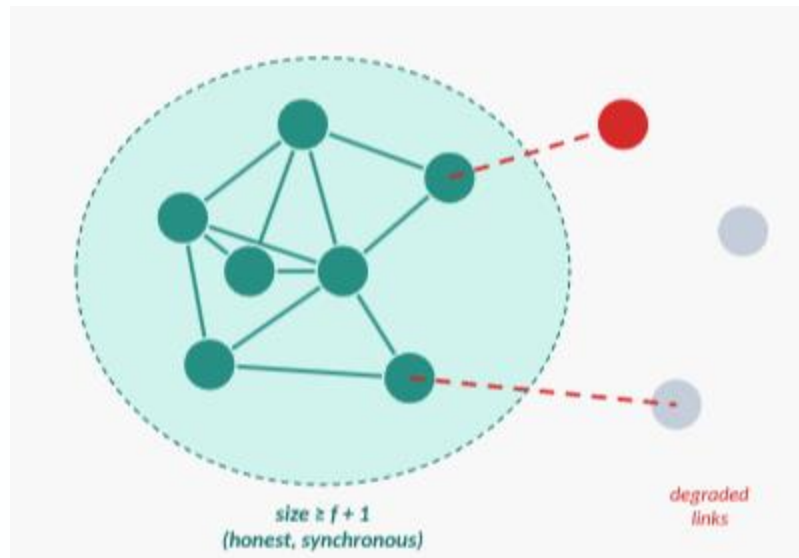


- An adaptive adversary chooses f Byzantine nodes and up to β degraded links anywhere in the network.
- If budget $\beta < h(n-2f-1)$, a component of size $> f+1$ is guaranteed to survive. So, a constant degree expander will absorb a budget linear to network size.
- The unconstrained adversary can force up to a $O(\beta)$ fold multiplicative blow up of the original diameter and protocol runtime, by cutting the β shortest bottleneck paths.



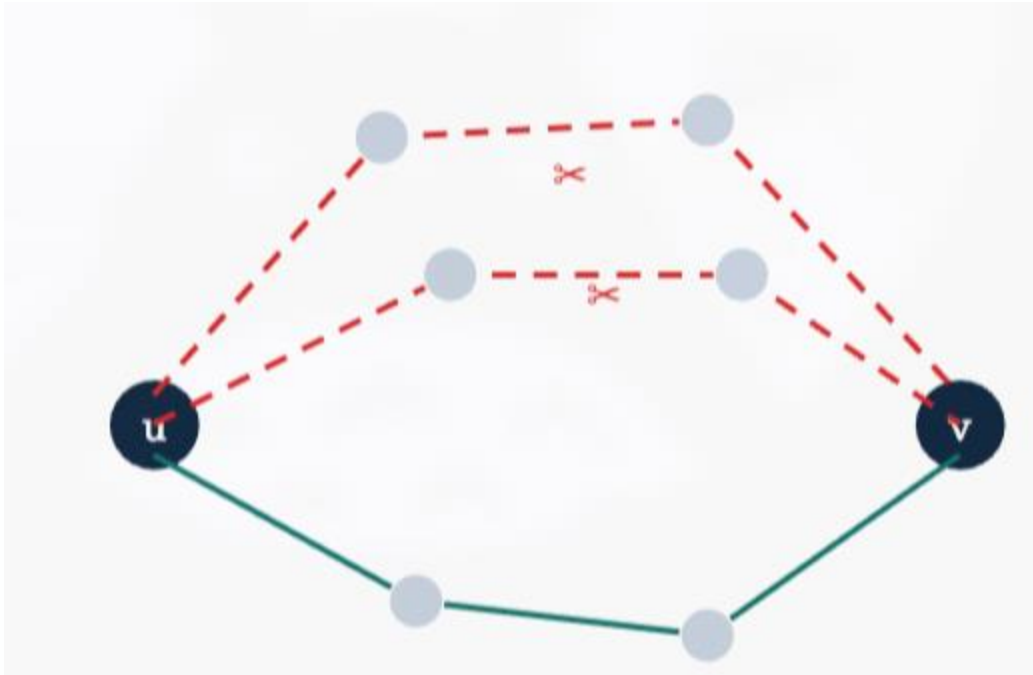
- The β degraded links must form a **connected subgraph**.
- The blowup to protocol runtime is now at most an additive $O(\beta)$.

When does consensus survive?



- Condition 1: $\kappa(G) = f + \beta + 1$
 - Proof idea: Replace edge deletions with vertex deletions, then no synchronous paths with a connectivity of $< f + \beta$ would violate the definition of connectivity. Satisfied since $n - f > f + 1$ in original consensus and all honest nodes must form a connected component.
- Condition 2: $\beta < h(n - 2f - 1)$
 - Proof idea: Splitting off any set of nodes S requires $h|S|$ edges and adversary has β edges to cut then if components must stay connected we set $|S|$ to $n - 2f - 1$ as $(n - f) - (n - 2f - 1) = f + 1$

How much do the timers slow down?



- Timers are proportional to the diameter of the graph, the unconstrained adversary can slow down the protocol roughly β times.
 - Proof idea: Suppose there are $\beta+1$ edge disjoint paths between any 2 honest links. Since adversary can degrade β paths at least one must survive. So diameter is bounded by longest diameter in graph where there are $(\beta+1)$ edge disjoint paths.

The constrained case advantage

- When does consensus survive?
 - Only one condition: $\beta < h(n - 2f - 1)$
 - Proof idea: since the edges are contiguous now, we must ensure no single ball has a boundary which can be severed by beta cuts so the edge expansion idea from there.
- How much do timers slow down?
 - Timers scale according to $d(\beta) < d + 2\beta + \phi(\beta)$
 - Proof idea: we route around the boundary of the malicious node ball.

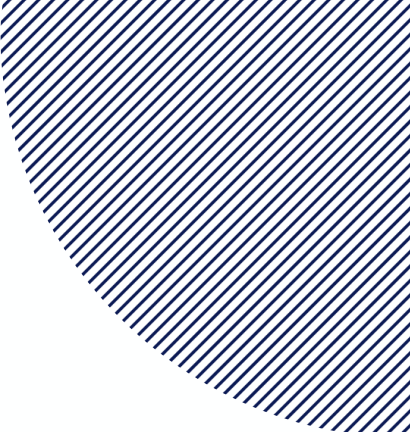


THANK YOU!

Questions?

Department / Unit





THANK YOU!

I ILLINOIS

Department / Unit